

UNIT 7: GRAPH OPERATIONS AND TREES

Operations, Tree Properties, and Computational Analysis

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2026-II



Course Outline

- 1 Operations on Graphs
- 2 Trees as Graphs
- 3 Algorithms and Analysis
- 4 Summary



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Basic Graph Operations

Set-Theoretic Operations

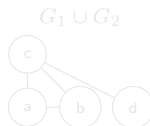
Graph operations **manipulate** graph structure through *set operations* on vertices and edges.

Union

Combine two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ into $G = (V_1 \cup V_2, E_1 \cup E_2)$. **Result:** All vertices and edges from *both* graphs.

Intersection

Retain **only common** vertices and edges: $G = (V_1 \cap V_2, E_1 \cap E_2)$. **Result:** *Shared subgraph* structure.



Basic Graph Operations

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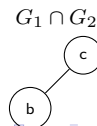
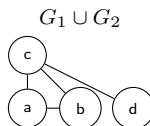
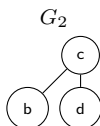
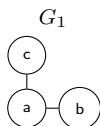
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Advanced Graph Operations

Ring Sum (*Symmetric Difference*)

Combine edges **not shared** by both graphs: $E = (E_1 \cup E_2) \setminus (E_1 \cap E_2)$.

Vertex and Edge Deletion

- **Delete vertex** v : **Remove** v and *all incident edges*.
- **Delete edge** e : **Remove** edge while preserving vertices.

Vertex Fusion

Merge two vertices u and v into a single vertex.



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Introduction to Trees

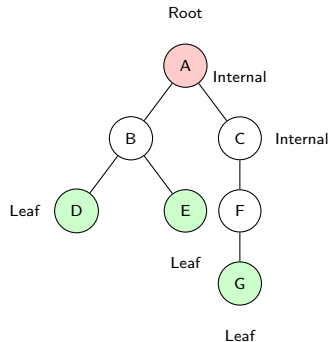
Definition

A **tree** is a *connected acyclic graph*. Key properties:

- $|E| = |V| - 1$
- **Unique path** between any two vertices
- *Minimum connected graph* structure

Tree Structure

- **Root:** Designated starting vertex
- **Leaves:** Vertices with degree 1
- **Internal nodes:** Vertices with degree ≥ 2



Distance Metrics in Trees

Distance between Vertices

The **distance** $d(u, v)$ is the *length of the unique path* from u to v .

Property: In trees, distance is **well-defined and unique**.

Eccentricity

The **eccentricity** $e(v)$ of a vertex v :

$$e(v) = \max_{u \in V} d(v, u)$$

Radius and Diameter

- **Radius:** $r(G) = \min_v e(v)$
- **Diameter:** $d(G) = \max_v e(v)$



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Graph Center

Definition

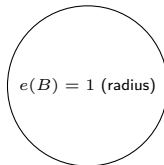
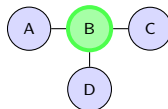
The **graph center** is the set of vertices with **minimum eccentricity**:

$$C(G) = \{v \in V \mid e(v) = r(G)\}$$

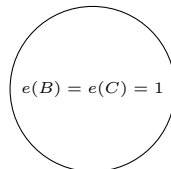
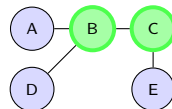
Properties

- 1 or 2 centers per tree
- If 2: they are *adjacent*
- Minimize max eccentricity

1 Center



2 Centers



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Algorithms for Graph Operations

Operation Complexity

Operation	Time	Space
Union	$O(V + E)$	$O(V + E)$
Intersection	$O(V + E)$	$O(V + E)$
Vertex deletion	$O(E)$	$O(1)$
Edge deletion	$O(1)$	$O(1)$
Vertex fusion	$O(E)$	$O(1)$

Distance Calculation Algorithm

- BFS/DFS from vertex v
- Track maximum distance encountered
- Return eccentricity $e(v)$
- Complexity: $O(|V| + |E|)$ per vertex

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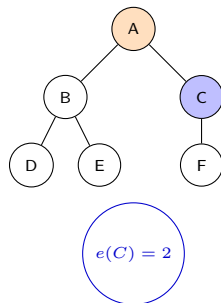
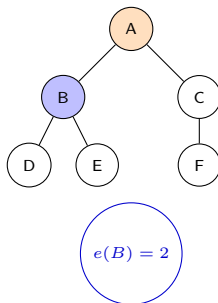
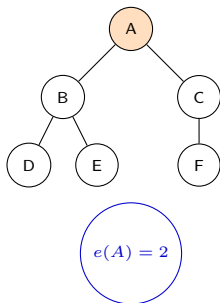
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Calculating Eccentricity: *Step-by-Step*

Compute $e(v)$ for each vertex



Key Insight

Eccentricity is the **maximum distance** from v to any other vertex. For each starting vertex, run BFS and track the farthest vertex reached.



Algorithm Pseudocode

Eccentricity

```

function Eccentricity( $G, v$ )
   $dist \leftarrow \text{BFS}(G, v)$ 
   $max\_dist \leftarrow 0$ 
  for each  $u \in V$  do
    if  $dist[u] > max\_dist$ 
       $max\_dist \leftarrow dist[u]$ 
    end if
  end for
  return  $max\_dist$ 
end function

```

All Eccentricities

```

function AllEccentricities( $G$ )
   $ecc \leftarrow []$ 
  for each  $v \in V$  do
     $ecc[v] \leftarrow$ 
      Eccentricity( $G, v$ )
  end for
  return  $ecc$ 
end function

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Optimization Strategies

Distance Matrix Caching

For **repeated queries**, precompute and cache all pairwise distances. Trade memory for speed: $O(|V|^2)$ space for $O(1)$ query time (Floyd-Warshall preprocessing).

Tree Property Exploitation

Since trees have **unique paths**, use **LCA (Lowest Common Ancestor)** techniques:

- Precompute binary lifting in $O(|V| \log |V|)$
- Query distance in $O(\log |V|)$ time
- Useful for **many queries** on static trees



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Finding Tree Centers

Algorithm Overview

Leaf Stripping Method:

- 1 Start with **all vertices** as candidates
- 2 *Iteratively remove* leaves (degree 1 vertices)
- 3 When $|V| \leq 2$ vertices remain: **these are the centers**

Complexity

Time: $O(|V| + |E|)$ **Space:** $O(|V|)$

Alternative Approach

Find vertex with **minimum eccentricity** through *diameter endpoints* (two BFS passes).



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Complexity Analysis Framework

Factors Affecting Performance

- **Graph density:** *Sparse vs. dense* affects algorithm selection
- **Representation:** Adjacency list vs. matrix impacts operation costs
- **Operation frequency:** *Trade-off* between preprocessing and query time
- **Memory constraints:** *Space-time trade-off* in practical systems

Best Practices

- Use adjacency lists for *sparse trees*
- Cache computed distances when repeatedly queried
- Leverage tree structure (*unique paths*) for optimizations



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Topics Covered

✓ Graph Operations

Union, Intersection, Ring sum, and *structural modifications*.

✓ Trees as Graphs

Specialized graph structures with *unique properties*.

✓ Distance, Eccentricity, and Center

Metrics for analyzing *tree topology* and finding *central nodes*.

✓ Algorithms and Analysis

Efficient computational methods and *complexity characterization*.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/computer-sciences-ii>

